

Event Bubbling and Delegation in JavaScript

Understanding **event bubbling** and **event delegation** is essential for writing clean, efficient event-driven code in JavaScript.

What is Event Bubbling?

When an event occurs on a DOM element, it **bubbles up** through its ancestors. This means the event is first captured and handled by the **target element**, and then propagated upward to its parent, grandparent, and so on.

Example:

```
<div id="parent">
  <button id="child">Click Me</button>
</div>
```

```
document.getElementById('child').addEventListener('click', () => {
  console.log('Child clicked');
});

document.getElementById('parent').addEventListener('click', () => {
  console.log('Parent clicked');
});
```

Output when button is clicked:

```
Child clicked
Parent clicked
```

This shows that the event bubbles from the child to the parent.

Stopping Event Bubbling

To stop the event from bubbling up the DOM tree, use:

```
event.stopPropagation();
```

```
button.addEventListener('click', (event) => {  
  event.stopPropagation();  
  console.log('This won't bubble up');  
});
```

What is Event Delegation?

Event delegation is a technique where a single event listener is added to a **common parent** instead of individual child elements. This works because of event bubbling.

Why Use Event Delegation?

- Improved performance with many child elements
- Simplified code
- Useful for dynamic elements added after page load

Example:

```
<ul id="menu">  
  <li>Home</li>  
  <li>About</li>  
  <li>Contact</li>  
</ul>
```

Instead of attaching a click listener to every `li`, delegate it to the `ul`:

```
document.getElementById('menu').addEventListener('click', (event) => {  
  if (event.target.tagName === 'LI') {
```

```
    console.log(`You clicked on ${event.target.textContent}`);  
  }  
});
```

This even works if new `<li>` elements are added later using JavaScript.

Summary

- **Event bubbling:** Events move up the DOM tree from child to parent.
- Use `event.stopPropagation()` to prevent bubbling.
- **Event delegation:** Handle events at a parent level for better performance and maintainability.